

AWS Cost Optimization Strategies

Accelerate business efficiency, eliminate idle resources, and maximize return on cloud investments.

⚙️ Right-Sizing

⚡ Spot & Savings Plans

📁 S3 Lifecycles

📈 FinOps Governance

| Right-Sizing Compute Resources

⚠ The Over-Provisioning Pitfall

Deploying instances larger than required leads to wasted allocation. Most workloads operate under 15% average CPU utilization.

- ✓ **AWS Compute Optimizer:** Leverage machine learning algorithms to map historical traffic and workload profiles to the ideal EC2 shapes.
- ✓ **Scale Vertically & Horizontally:** Reduce direct instance size or use Auto Scaling to turn off non-production tiers during idle hours.

STANDARD CONFIGURATION

t3.large

\$60 / Month



RIGHT-SIZED CONFIGURATION

t3.medium

50% Monthly Savings (\$30)



| Leveraging **Spot Instances & Savings Plans**

Dynamic Compute Pricing

On-Demand instances offer flexibility but represent the absolute ceiling of cloud pricing. Selecting the correct commitment models guarantees reduction.

Spot Instances

Use spare AWS capacity for fault-tolerant and stateless applications (e.g., CI/CD, batch processing) for **up to 90% savings**.

Savings Plans

Commit to consistent hourly compute usage for 1 or 3 years. This applies globally to EC2, Fargate, and Lambda for **up to 72% savings**.

AWS Pricing Model Cost Comparison

On-Demand

Baseline Cost (100%)

Savings Plans

28% to 45% (Savings up to 72%)

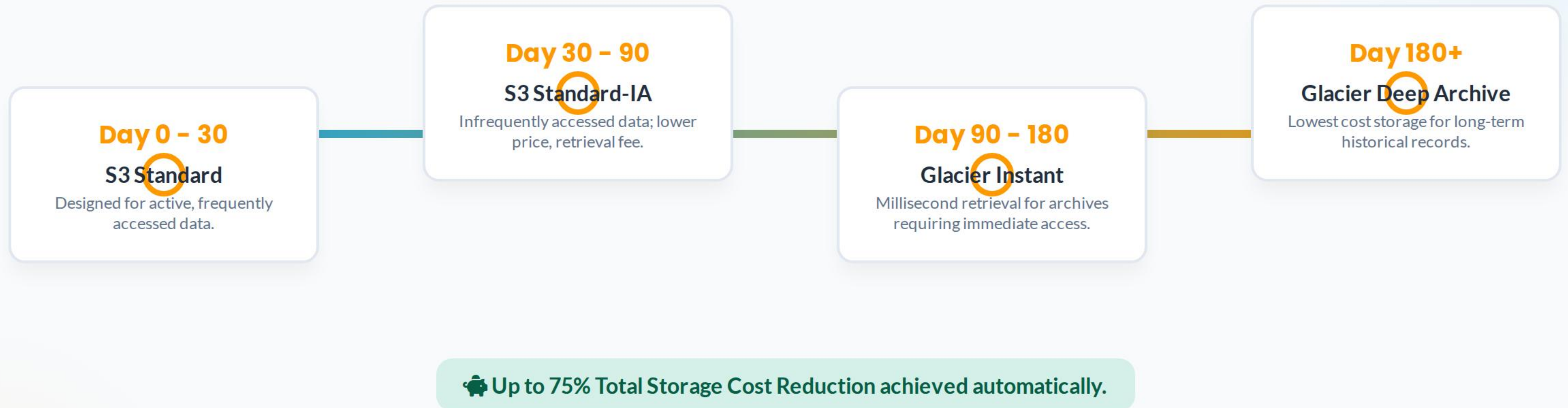
Spot Instances

up to 90% savings

*Actual savings depend on instance families and location commitment models.

| S3 Storage Lifecycle Policies

Indefinite storage in S3 Standard is an expensive strategy for backups, logs, and compliance records. Automated lifecycles transition files to cheaper tiers as they age.



The AWS Cost Optimization **Toolbox**

Compute Services

Define, schedule and scale compute capacity automatically based on real-time trends.

Services: Amazon EC2, AWS Lambda, Amazon ECS

Intelligent Storage

Store objects in performance-optimized layers without administrative overhead.

Services: Amazon S3, S3 Intelligent-Tiering, EFS Lifecycle

Analytics & Governance

Analyze trends, allocate budgets, construct anomaly detection alerts and map costs.

Services: AWS Cost Explorer, Budgets, Compute Optimizer



1. Monitor
(CloudWatch)



2. Analyze (Cost
Explorer)



3. Recommend
(Optimizer)



4. Execute (Auto
Scaling)

| Real-World Business Impact

50%

Compute Sizing

Reduction in compute overhead via direct right-sizing adjustments.

90%

Spot Workloads

Savings achieved for test networks & continuous delivery pipelines.

75%

Storage Archive

Savings by transitioning S3 Standard to cold Glacier tiers.

35%

Overall Reduction

Typical average bill decrease in the first 90 days of execution.

Executive Outcomes

- ✓ **Preserve High Performance:** Right-sizing ensures workloads stay responsive while cutting away idle capacity.
- ✓ **Reallocate Cloud Budgets:** Shift finances from maintenance costs to feature engineering and product scaling.
- ✓ **Eliminate Manual Governance:** Implement policies once, and let automation preserve margins.



| Continuous **FinOps Lifecycle**

Cost optimization is not a one-time project. It is a recurring management practice requiring architectural governance and team accountability.

Lifecycle Phase	Core Activities & Frameworks	Direct Architectural Benefit
1. Inform & Analyze	Deploy tag governance and allocate costs directly to individual teams or services. Identify trends.	Complete visibility into business cost drivers and exact ownership.
2. Optimize	Implement Compute Optimizer recommendations, upgrade families (e.g., GP2 to GP3 storage), size instances.	Immediate removal of wasted cloud allocation without sacrificing up-time.
3. Operate	Enable automated scaling policies, S3 Lifecycle policies, and buy Reservation models dynamically.	Long-term price security and predictable infrastructure overhead.



Achieve **Operational Excellence**

"Cost optimization is a continuous process that helps organizations maximize cloud efficiency while maintaining performance, availability, and scalability."

✓ Compute Optimized

✓ Storage Modernized

✓ Budgets Governed

| Image Sources



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